

Shivani Kalal

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AI/ML Engineer with 2+ years building production agentic RAG systems, LLM evaluation frameworks, and deep learning models – from transformer architectures built from scratch to end-to-end GenAI pipelines. Open-source contributor with hands-on experience in model evaluation, AI safety, and inference optimization. Skilled in PyTorch, LangChain, LangGraph, and OpenAI/Claude APIs; MS Computer Science (GPA 3.76).

Technical Skills

Languages: Python, Java, TypeScript

LLMs & Agentic AI: OpenAI GPT, Anthropic Claude, LangChain, LangGraph, agentic workflows, multi-layer RAG, prompt engineering, LLM evaluation, AI safety

AI & Machine Learning: Deep Learning, NLP, Transformers, RAG, semantic search, vector & hybrid retrieval, embeddings, model training & evaluation, model deployment, MLOps

Model Optimization & Evaluation: ablation studies, hyperparameter tuning, benchmarking, evaluation framework design, systematic model comparison

Frameworks & Libraries: PyTorch, Scikit-learn, NumPy, Pandas, Django, Flask, Spring Boot, spaCy, Streamlit

Databases & Cloud: ChromaDB, FAISS, PostgreSQL, MySQL, MongoDB, SQLite, AWS, Azure

DevOps & Tools: Docker, CI/CD, Azure DevOps, Databricks, Kafka, Git, GitHub, Linux/Unix

AI Dev Tools: Claude Code, Cursor, Copilot

Frontend & Web: React.js, Node.js, TypeScript, HTML/CSS

Engineering Practices: REST APIs, OOP, System Design, DSA, Unit Testing, Agile/Scrum, Jira

Professional Experience

Software Engineer, *Value Momentum Software Services, Hyderabad* Jun 2022 – Jul 2024

- Architected and deployed a production LLM evaluation framework with automated model evaluation, structured-output validation, prompt experimentation, and inference-time guardrails to continuously measure model quality and reliability, reducing invalid generations by 35% across AI workflows serving 500+ users at 15K–20K daily requests.
- Developed and deployed a production LLM application using OpenAI models to convert natural-language requirements into structured Gherkin specifications, generating 4,000+ automated test cases adopted by 10+ engineers while eliminating manual test authoring across the platform.
- Built CI/CD and MLOps pipelines in Azure DevOps for 20+ Databricks training and inference workflows, and instrumented production observability – inference latency, token usage, output-quality metrics, model drift, and cost trends – across LLM workflows, cutting deployment time 30% and mean-time-to-detect production issues 25%.
- Engineered scalable data validation and preprocessing pipelines in Azure Databricks, improving feature quality, reducing downstream pipeline failures by 30%, and ensuring reliable inputs for production analytics and AI workloads processing over 2M daily records.
- Designed asynchronous AI workflow orchestration with retry handling, queue-based execution, idempotent request processing, and resilient API services to support reliable, high-volume inference workloads while optimizing API latency by 15%.

Software Developer, *University of Mississippi, Oxford* Aug 2024 – Jul 2025

- Architected a full-stack research repository for 500+ researchers (React, Node.js, TypeScript, PostgreSQL) with metadata-driven document management, advanced search, and role-based access control across 20+ relational tables managing 30,000+ plant records under a 7-level taxonomical classification.
- Optimized system performance via PostgreSQL indexing and Redis caching, cutting query latency from 400ms to 250ms and eliminating 60% of redundant queries; improved frontend load times 44% through lazy loading, code splitting, and React memoization.

Graduate Teaching Assistant, *University of Mississippi, Oxford* Jan 2025 – May 2026

- Tutored 45+ students across 3 Computer Networks sections on debugging, programming assignments, and TCP/IP labs, translating complex technical concepts for non-expert audiences and lifting assessment scores 40%.

Projects

MARS: Memory-Augmented Research System | GitHub | *Python, LangChain, BM25, ChromaDB* Jan 2026

- Built and open-sourced a production-grade agentic research system with multi-step workflows, a 3-layer memory architecture (episodic auto-summarization, cross-session ChromaDB + SQLite retrieval), and a hybrid BM25 + MPNet semantic retrieval pipeline; designed a standalone evaluation framework achieving 95% citation verification and 2x source coverage, outperforming SOTA tools Consensus (57% vs. 31%) and Elicit across all benchmarks.

RagShield: LLM Security Framework | GitHub | *Python, LangChain, ChromaDB* May 2026

- Built and open-sourced an empirical AI safety framework for production agentic RAG systems, executing 2 indirect prompt-injection attacks, a known LLM attack vector, that reached 100% attack success rate by slipping fabricated claims past automated QA, then engineering 2 prompt-level defenses and monitoring checks that cut attack success rate to 0% with zero utility loss, validated through a 3-condition ablation study.

LangFormer: Transformer for Neural Machine Translation | GitHub | *Python, PyTorch* Mar 2025

- Built a Transformer-based NMT model entirely from scratch in PyTorch, implementing multi-head self-attention, positional encodings, residual connections, and beam-search decoding. Conducted a systematic hyperparameter search across four configurations (embedding dimension, hidden size, dropout, and learning rate), achieving a 22% BLEU improvement over the Seq2Seq baseline and deploying inference through both CLI and REST APIs.

Education

Master of Science, Computer Science, *University of Mississippi, Oxford* | GPA: 3.76 Aug 2024 – May 2026

Relevant Courses: Deep Learning, Computer Vision, LLM Security, Data Mining, Analysis of Algorithms, Computer Networks